

Master's Dissertation – Will you take up the challenge?

Module 3 – Writing of Dissertation and Extended Abstract

Ana Carvalho (Dept. Engineering and Management, DEG),

Beatriz Silva (Dept. Mechanical Engineering, DEM),

Isabel Gonçalves (Tutoring Office, GATu),

João Ramôa Correia (Dept. of Civil Engineering, Architecture and Georesources, DECivil)

Pedagogical Council

Will you take up the challenge?

1 Σ Emotional
and
Interpersonal
Aspects



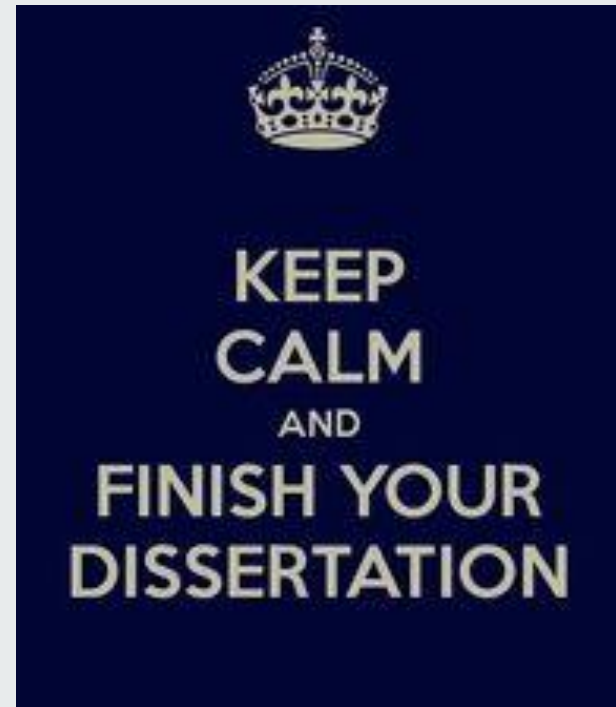
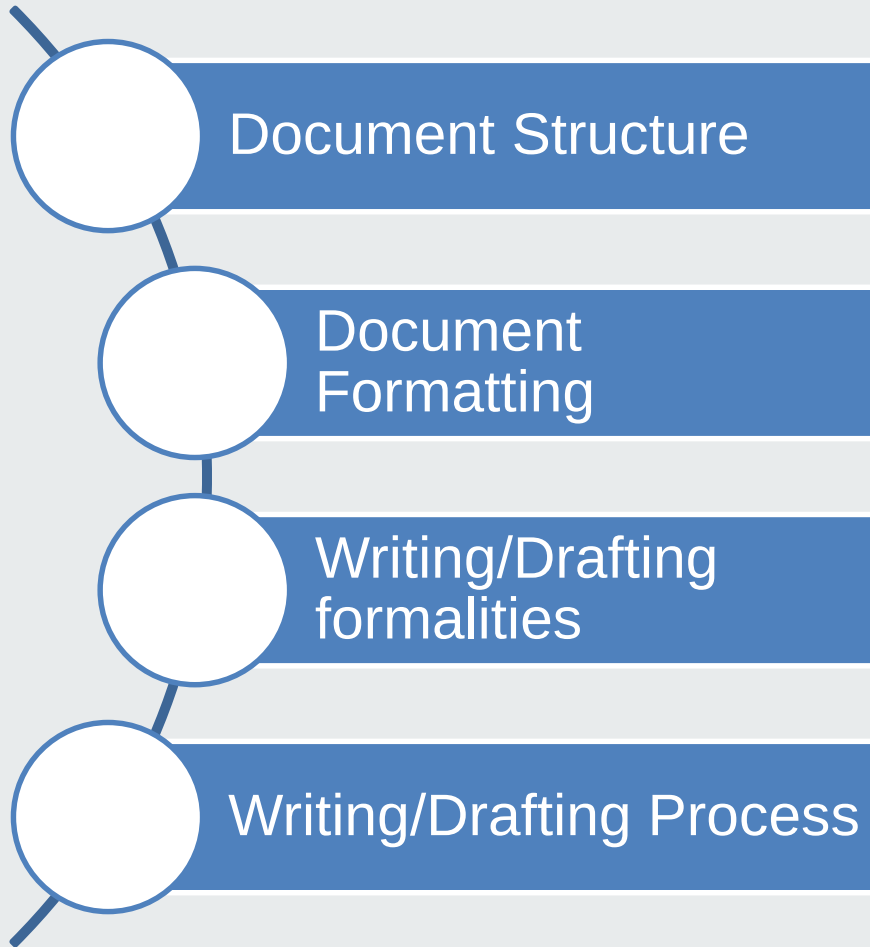
2 Σ Working
Methodology



3 Σ Writing of
Dissertation
and
Extended
Abstract

4 Σ Presentation
and defense

Writing of Dissertation and Extended Abstract



What should a dissertation include?

What was done and why?



Introduction / Objectives

What is known? And what is unknown?



Literature review

How was it done?



Methodology

What were the findings?



Results

What do the results mean? Benchmarking with state-of-the-art? What are the possible implications and recommendations?



Discussion

What are the main results?
What are the recommendations for the future?



Conclusion and Future Work

Dissertation Structure

Dissertation

Pre-text elements

Cover

Dedicatory

Abstract and keywords

Acknowledgments

Table of contents

Other tables of contents / lists

Text elements

Introduction

State-of-the-Art

Methods

Results and Discussion

Conclusion

Post-text elements

References

Annexes

Without
numbering

Page
number
formatting
i, ii, ...

Numbering
1, 2, ...

Page
number
formatting
1, 2, ...

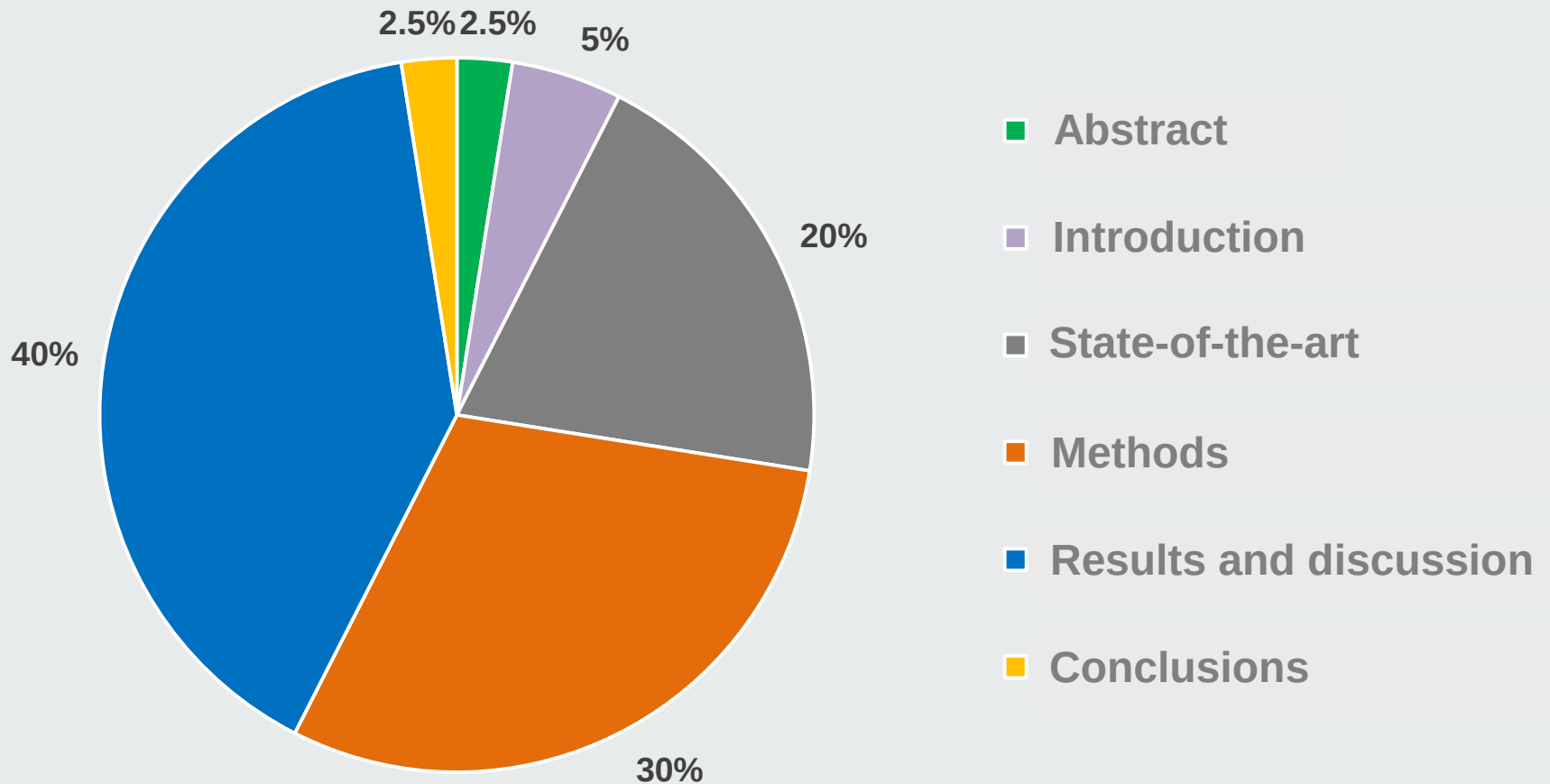
A, B, ...

A1, A2, ...

Mandatory

Optional

Relative Weight of Chapters

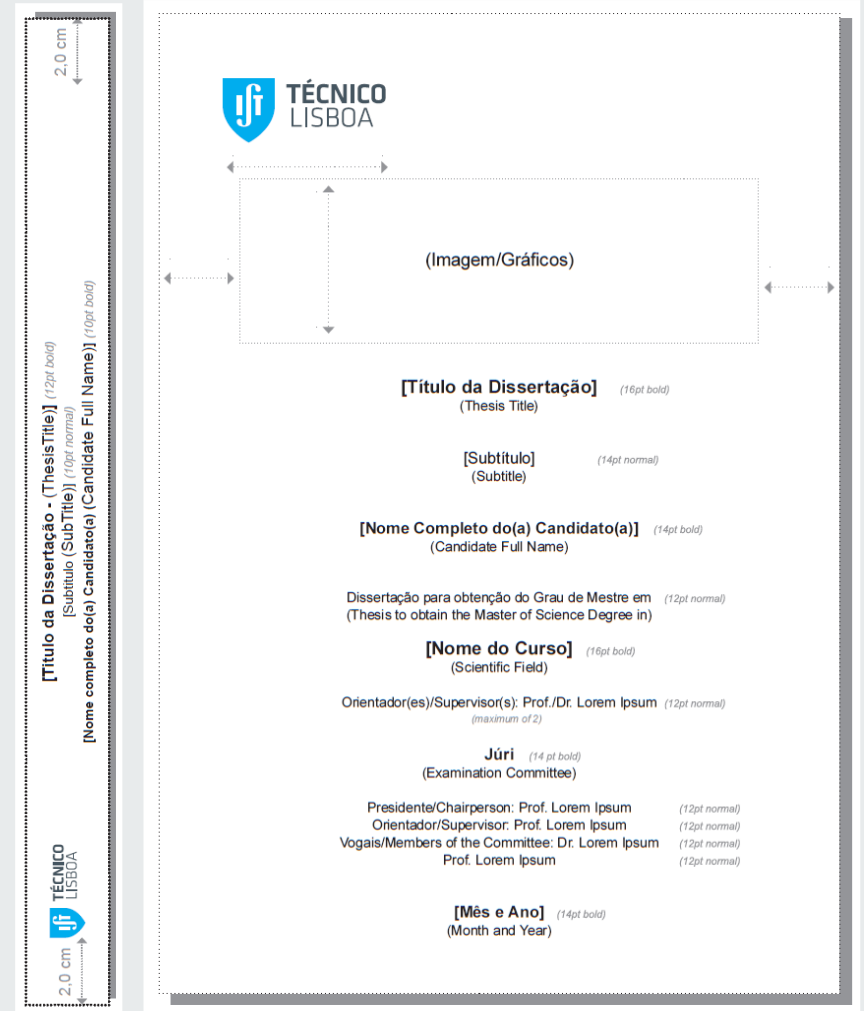


Dissertation Structure

Cover and spine

The cover of a dissertation identifies the title, the author, the institution and the date of conclusion.

The title of the dissertation describes logically, carefully and shortly the core issue of the document.



Spine (Left):

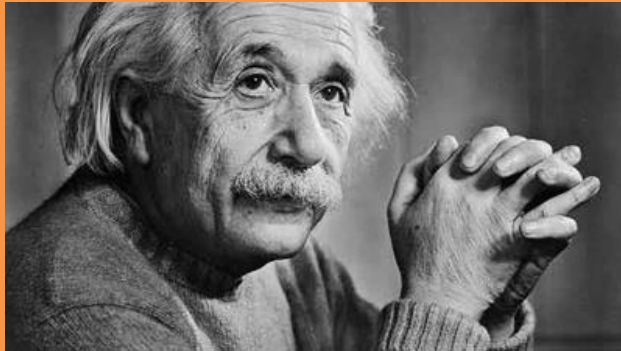
- [Título da Dissertação - (Thesis Title)] (12pt bold)
- [Subtítulo (SubTitle)] (10pt normal)
- [Nome completo de(a) Candidato(a) (Candidate Full Name)] (10pt bold)
- TÉCNICO LISBOA

Cover (Right):

- TÉCNICO LISBOA
- (Imagem/Gráficos)
- [Título da Dissertação] (16pt bold)
(Thesis Title)
- [Subtítulo] (14pt normal)
(Subtitle)
- [Nome Completo do(a) Candidato(a)] (14pt bold)
(Candidate Full Name)
- Dissertação para obtenção do Grau de Mestre em (12pt normal)
(Thesis to obtain the Master of Science Degree in)
- [Nome do Curso] (16pt bold)
(Scientific Field)
- Orientador(es)/Supervisor(s): Prof./Dr. Lorem Ipsum (12pt normal)
(maximum of 2)
- Júri (14 pt bold)
(Examination Committee)
- Presidente/Chairperson: Prof. Lorem Ipsum (12pt normal)
- Orientador/Supervisor: Prof. Lorem Ipsum (12pt normal)
- Vogais/Members of the Committee: Dr. Lorem Ipsum (12pt normal)
- Prof. Lorem Ipsum (12pt normal)
- [Mês e Ano] (14pt bold)
(Month and Year)

Dissertation Structure

Dedicatory



Acknowledgments



Optional, but can freely express their views

Dissertation Structure

Must contain the thesis summary

Resumo

A globalização da economia tem levado não só a uma aceleração da internacionalização das empresas, como também a uma maior prática do *outsourcing* logístico. Neste contexto, o transporte rodoviário de mercadorias assume particular importância, pois as alterações que se verificaram no mercado exigem maiores distâncias de transporte para a distribuição dos produtos e, consequentemente, maiores custos de transporte que representam uma parte significativa dos custos logísticos. Na contratação de parceiros logísticos, as tarifas cobradas pelos transportadores na prestação de serviços são um ponto de discussão no processo de negociação entre as partes.

Assim, num ambiente de maior competitividade, as empresas expedidoras estão cada vez mais determinadas a compreender as tarifas de mercado cobradas pelos seus operadores logísticos, no sentido de conseguirem racionalizar os seus custos. O presente trabalho surge como uma possível ferramenta para ajudar na resposta a esta necessidade.

Em primeiro lugar, focou-se na compreensão clara do problema em questão e na revisão bibliográfica de alguns conceitos relevantes nesta matéria. Seguidamente é apresentada uma estrutura teórica de custos de transporte. O estudo realizado incide sobre a análise das componentes de uma estrutura de custos do transporte de mercadorias e dos factores a que esta é sensível. O resultado deste estudo pretende servir de instrumento auxiliar de referência, com fundamentação teórica, na negociação das tarifas anuais dos contratos de prestação de serviços entre operadores logísticos e as empresas expedidoras, visando colmatar uma falha actualmente existente na definição dos critérios para as variações tarifárias dos transportes. Esta será a base para o desenvolvimento de um modelo de apoio à tomada de decisão a desenvolver em trabalho futuro na Dissertação.

Palavras-chave: Custo, *Outsourcing* logístico, Transporte, Tarifa

Problem presentation and methodology

Problem contextualization

Main results obtained

Keywords characterize the document field of action (4 to 6)

Dissertation Structure

Índice

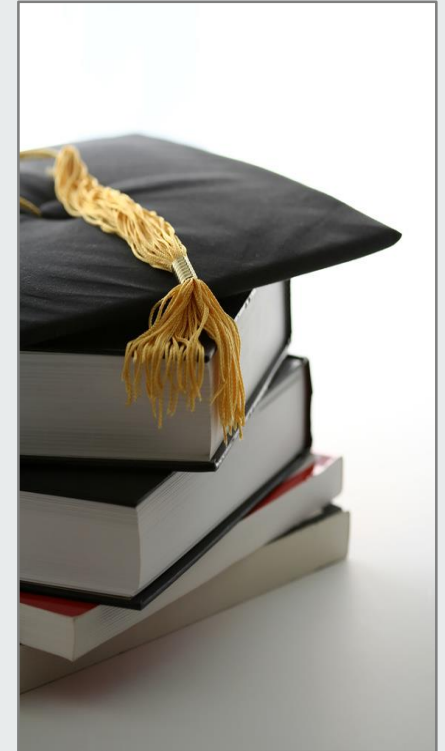
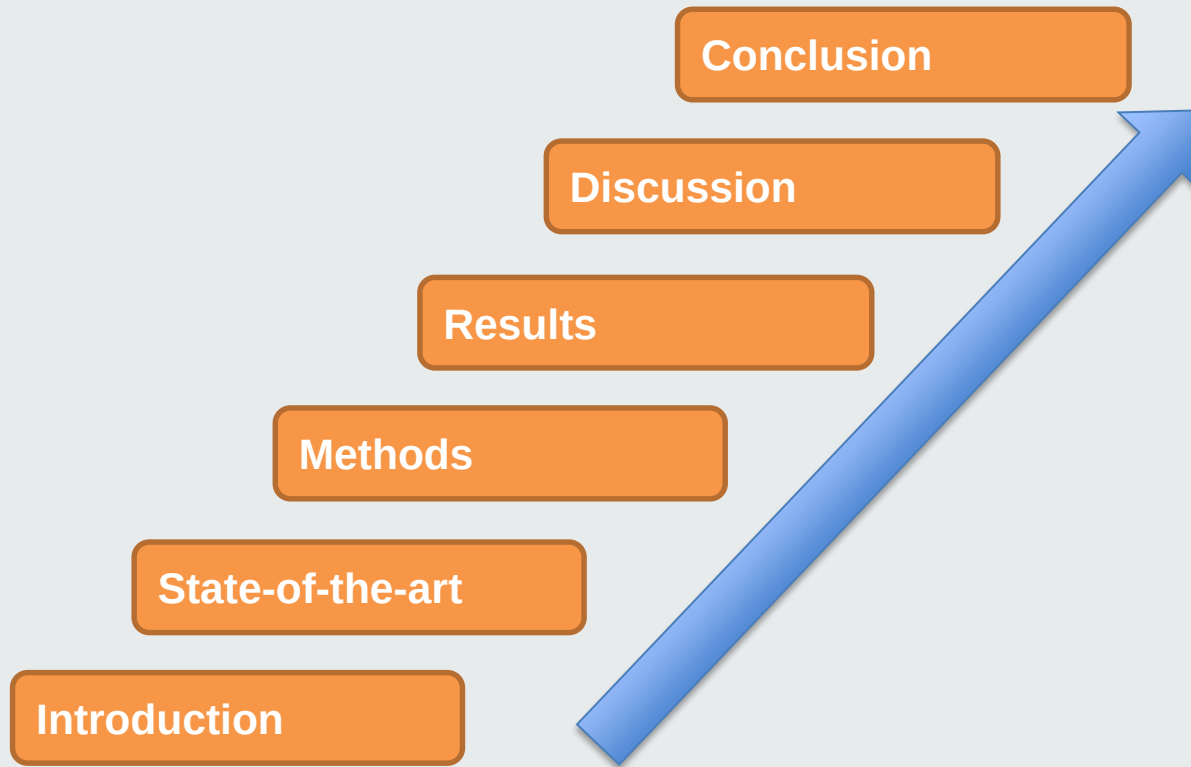
Agradecimentos	ii
Resumo.....	iii
Abstract.....	iv
Índice.....	v
Lista de Abreviaturas.....	vii
Lista de Índices	viii
Lista de Figuras	viii
Lista de Tabelas.....	ix
1 Introdução	1
1.1 Contextualização do Problema	1
1.2 Metodologia.....	1
1.3 Objectivos do Trabalho	2
1.4 Estrutura da Dissertação	3

Figure Index + Table Index + List of Symbols and Acronyms
+ Software

A good structure is crucial to have a good
“story”



Dissertation Structure



A thesis should be a “story” and have a good underlying theme.
The story must be “told” considering what the reader (already) knows in a given moment.

GENERAL CONSIDERATIONS



Document body, 80 pages max (from introduction to conclusions)



Annexes, 20 pages max



Balanced number of pages among chapters



5 to 7 chapters recommended



4 section levels max (4.; 4.1; 4.1.1; 4.1.1.1); 3 levels recommended; at least 2 sub-chapters per section

GENERAL CONSIDERATIONS



Use spell checker (however it does not detect all mistakes...)



Do not split chapter titles / text sections, table or figure from the respective caption (Word functionality: *'keep with next'*)



Do not be fully dependent on word-processor automatic formatting; you can do almost everything “manually”



Use always the same font; figure and table captions use the same size of font, even though slightly smaller
Coherent text in figures and tables coherent with other formatting



Use the aggregate space between numbers and the respective unit to avoid splitting numbers when you change to a new line (Word functionality: *'Ctrl + Shift + Space'*)

GENERAL CONSIDERATIONS



Do not use capital letters unnecessarily



Ensure that the text area is completely filled. This may imply breaking some formatting rules (e.g.: one figure or table must only appear on the text after the respective reference; one figure or table must be included in the respective sub-chapter / section)

Suggestion: check the final version only (before making a pdf and printing)



Respect margins.

Document formatting

Font type: Arial (or similar)

Font size: 10

Margins: 2,5 cm, min, equally spaced, all around

Resumo

Actualmente, as organizações competem num mercado global e exigente, que as força a avaliar constantemente o seu desempenho em termos de eficiência e eficácia na obtenção dos seus objectivos. Apesar da constante evolução nesta área, continua ainda a existir alguma dificuldade na definição da melhor forma de avaliação de desempenho, que é geralmente conseguida através da definição de indicadores. Esta tarefa é muitas vezes uma tarefa difícil, consumindo inúmeros recursos. A escolha dos indicadores de performance e sua integração num sistema de avaliação de desempenho, assume-se como um desafio para os gestores do século XXI, pois permitirá acompanhar e melhorar o desempenho global de uma organização.

É neste contexto, que surge o problema em estudo, da fábrica da Volkswagen Autoeuropa, maior produtor e exportador nacional do ramo da indústria automóvel. A empresa pretende criar um sistema de acompanhamento dos objectivos operacionais da logística, com o intuito de avaliar o desempenho das diferentes áreas da logística. O sistema terá como base um conjunto de KPIs (*Key Performance Indicators*) que devem ser definidos e adequados a cada área. Este problema justifica e motiva a realização da presente dissertação em Engenharia e Gestão Industrial.

Numa primeira fase, o presente trabalho descreve o problema em análise que se enquadra no caso de estudo real da Autoeuropa. Posteriormente, é apresentada uma revisão bibliográfica sobre os temas de logística, análise e sistemas de desempenho e dashboards. Para resolução do problema, serão desenvolvidos KPIs logísticos e será implementado um *dashboard* de modo a acompanhar os objectivos operacionais da AE.

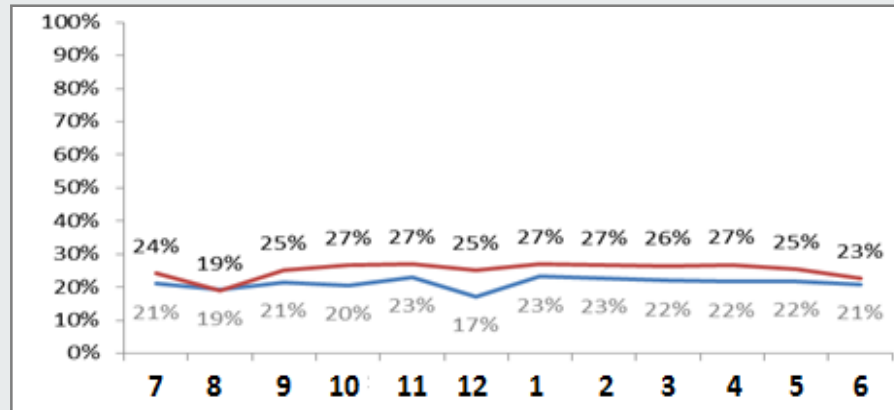
Este trabalho tem dois objectivos principais: (1) desenvolvimento de KPIs para a Logística da Autoeuropa e (2) criação de um sistema que permita o acompanhamento dos objectivos operacionais da Logística.

Palavras-chave: Logística, Avaliação de desempenho, Indicadores de Desempenho (KPIs), *dashboards*.

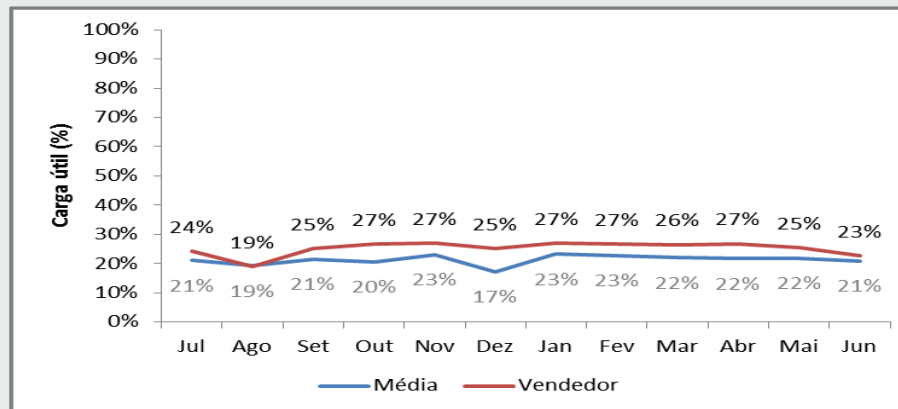
Spacing: 1,5

Page number: right-aligned

Figure 1 – Average monghtly vehicle overloads



WRONG



RIGHT

Figure 1 – Average monghtly vehicle overloads



Material	Ângulo com a direcção de laminagem	Módulo de elasticidade	Tensão de cedência	Tensão de rotura	Coefficiente de anisotropia	Alongamento após fractura
AISI304L	0	205852	273	650	0.69	40
	45	171513	288	585	1.06	49.1
	90	190203	316	626	0.70	66.8
	Média	184770	291	612	0.88	51.3

Tabela 3.2 – Propriedades mecânicas do aço AISI304L

WRONG



Tabela 3.2 – Propriedades mecânicas do aço AISI304L (Montanari, 2013b).

Material	Ângulo com a direcção de laminagem [°]	Módulo de elasticidade, E [MPa]	Tensão de cedência, σ_e [MPa]	Tensão de rotura, σ_R [MPa]	Coefficiente de anisotropia, r	Alongamento após fractura, A [%]
AISI304L	0	205852	273	650	0.69	40.0
	45	171513	288	585	1.06	49.1
	90	190203	316	626	0.70	66.8
	Média	184770	291	612	0.88	51.3

RIGHT

SYSTEMATIZING...



All figures and tables must be numbered and captioned (figure caption below, table caption above)



Use always the term 'figure' (irrespective of being a figure, photo, image or chart)



Use the term 'table' or 'illustration' (the former is preferred)



All figures and tables must be mentioned in the text and appear after the reference (except for exceptional situations)



Respect the margins and pay particular attention to figures and tables
If necessary, rotate table or figure and include it on a single page

SYSTEMATIZING...



When you include two or more figures side-by-side, it is easier to insert a table (no borders) and format the figures so that they have the same height



Figures must have a minimum quality / resolution



Figures and tables in general cannot be split in different pages. If it is not possible to fulfil this rule, the caption must be repeated and it should be clearly mentioned that it is the continuation of the figure / table



Reference to adapted figures (change of image or translation) must be “(adapted from...)”

Center equation

Introduce numbering of equation on the right side of the page

Esta abordagem pressupõe a existência de deformação plástica segundo trajectórias de deformação proporcionais definidas por

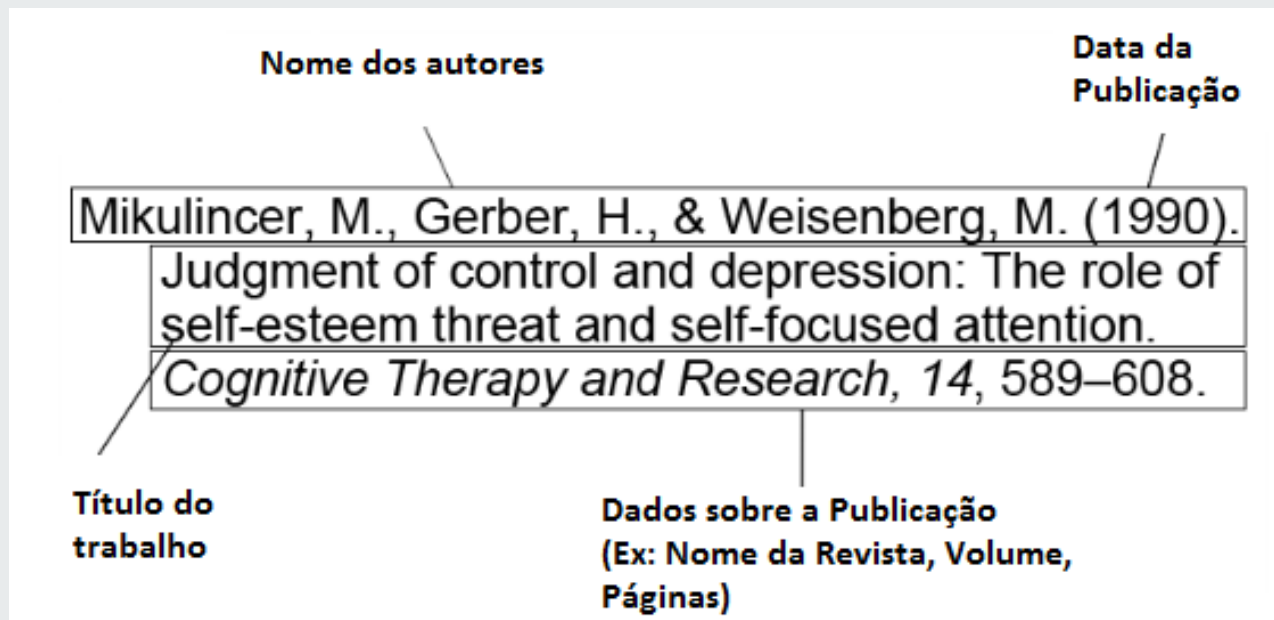
$$\beta = \frac{d\varepsilon_2}{d\varepsilon_1} = \frac{\varepsilon_2}{\varepsilon_1} \quad (3.5)$$

As extensões principais ($\varepsilon_1, \varepsilon_2$) são medidas a partir dos eixos maior e menor das elipses que resultam da deformação plástica dos círculos da grelha durante o fabrico das flanges (ver Equação (3.3)) e a extensão principal ε_3 na direcção da espessura, é obtida pela condição de incompressibilidade.

Define the meaning of all variables and parameters

REFERENCES

There are different reference formatting styles (Ex: APA style, Harvard referencing, etc.) – **THE IMPORTANT THING IS TO KEEP ALWAYS THE SAME STYLE**



DIRECT QUOTE

It is the true transcription of the document consulted. It should be used in a thoughtful/measured manner. The information must be quoted in quotation marks, and include the page.

Example: *definition of sustainability as “development that meets the needs of the present without compromising the ability of future generations to meet their own needs” (WCED, 1987, p. 24).*

INDIRECT QUOTE

It's the author's own interpretation regarding the information included in a document that is being cited. Explain the idea of another author through our own words.

Examples:

1) If you are making reference directly to the author:

Mont (2002) says that the services have developed in the manufacturing industry

2) If you express somebody's ideas

The supply chain performance is an increasingly important element for business competitiveness (Estampe et al., 2010).

For both types of quote you must be careful about **deontological issues (plagiarism)**.



QUOTE OF ANOTHER QUOTE

A document (which was not possible to consult) may still be quoted provided that it has been mentioned in a document that you have consulted (e.g.: a reviewed paper or a book), but this must be clearly acknowledged.

Example:

In the text: “Mont (2002), quoted by Faria (2004), refers that the services have developed inside the manufacturing industry.”

Both documents must be mentioned in the reference list. Regarding the document that was not directly consulted, at the end of the reference description, the document that quoted it must be mentioned: “(quoted by Faria, 2004)”.

QUOTE WITH AN AUTHOR

Mont (2002) or (Mont, 2002)

QUOTE W/ 2 AUTHORS

Ganga and Carpinetti (2011) or (Ganga and Carpinetti, 2011)

QUOTE W/ 3 OR MORE AUTHORS

Authors: Xiao, Y., Zhao, Q., Kaku, I., and Xu, Y

Quote: Xiao *et al.* (2012) or (Xiao *et al.*, 2012)

**QUOTE OF AN AUTHOR WHO PUBLISHES MORE THAN ONE
WORK PER YEAR (FORMAT: AUTHOR, YEAR)**

Mont (2002a), Mont (2002b), etc.

QUOTE W/ 1 AUTHOR

According to Mont [1]

QUOTE W/ 2 AUTHORS

According to Ganga and Carpinetti [2]

QUOTE W/ 3 OR MORE AUTHORS

Authors: Xiao, Y., Zhao, Q., Kaku, I., e Xu, Y

Quote: according to Xiao *et al.* [3]

QUOTE OF REFERENCES 1 AND 2

[1,2]

QUOTE OF REFERENCES 1, 2 AND 3

[1-3]

QUOTE OF REFERENCES 2, 3, 4 AND 8

[2-4,8]

REFERENCES

Identify all sources of information (books, magazines, monographs, reports, catalogues, internet, etc) where that information is actually quoted

References correspond to the works actually quoted in the text body; they should not include works consulted/studied but not quoted

References must be selected according to their effective contribution to the work

You must not exaggerate the number of references. You should only mention those that are actually relevant (do not “plant” references throughout the text - example: “*Previous studies [3-45] (...)*”)

In the list of references, the information supplied must be comprehensive (and consistent)

REFERENCES

DISSERTATION

Correia, J.R. (2008), GFRP Pultruded Profiles in Civil Engineering: Hybrid Solutions, Bonded Connections and Fire Behaviour. PhD Thesis in Civil Engineering, Instituto Superior Técnico, Universidade Técnica de Lisboa.

BOOK

Garner, G.O. (2003). Careers in Engineering (2nd ed.). Chicago, IL: VGM Career Books.

PAPER IN A SCIENTIFIC JOURNAL

Marvin, A. W., Schmidt, L. D., Benjaafar, S., Tiffany, D.G., & Daoutidis, P. (2012). Economic Optimization of a Lignocellulosic Biomass-to-Ethanol Supply Chain. *Chemical Engineering Science*, 67(1), 68–79. doi:10.1016/j.ces.2011.05.055

REFERENCES

COMMUNICATIONS IN CONFERENCE PROCEEDINGS

Silva, M. B., Skjoedt, M., Bay, N., Martins, P. A. F. (2011). Multistage single point incremental forming. CD-Rom Proceedings of the *Congress on Numerical Methods in Engineering*, 19 p.

SITES - INTERNET

In the text itself:

...the purchase cost of a hammer mill will be USD67.000, or approximately EUR52.000 (<http://www.xe.com/currencyconverter>).

In references:

A-Vision. (2014). Site da Autovision Portugal, <http://www.autovision.pt/>, viewed on 1 Janeiro.

HOW TO WRITE IT?



Use neutral form
(e.g: “An experimental work has been carried out...”)



Coherence of verbal forms in description (for example, do not use present and past simultaneously)



Coherence in decimal point separator, point or comma



Writing should be direct, objective and with a few adjectives



Be careful when you make considerations justifying all statements
Show and analyse information in all tables and figures

DO NOT USE COLLOQUIAL EXPRESSIONS

Came up

Increased

Came down

Decreased

Huge

Significantly



Extended abstract

The extended abstract has a 20% weight on the final thesis score

The structure should not be / should be the same the thesis structure

It should be the basis for a scientific paper

The paper body must include more on the work contribution

Written in English / do not translate word by word

10 pages max



KISS – Keep It Short and Simple

Structure of the extended abstract

Extended Abstract

Pre-text elements

Title

Author and affiliations

Abstract and keywords

Text Elements (paper body)

Introduction

Methods

Results and discussion

Conclusion

Numbering

1, 2, ...

Page numbers

1, 2, ...

Mandatory

Optional

Post-text elements

Acknowledgements

References

Title

- ✓ The title in English must not be translated directly (“literally”).

Abstract and keywords

- ✓ The Abstract must be **concise** and give an overall idea of the work carried out, how it was carried out and its main conclusions.
- ✓ In order to select the **keywords** imagine that you are looking for the paper in a database.



Introduction

- ✓ Introduce the topic and emphasize its importance
- ✓ Provide a framework with the work carried out previously, by identifying what has already been studied and what is necessary to study/carry out
- ✓ Introduce the work carried out, by identifying its purpose and major objective

Methods

- ✓ Describe the study object (*set-up* experimental, analytical, mathematical, computational, etc.)
- ✓ Explain the techniques used
- ✓ Be specific and supply all necessary data

Results and discussion

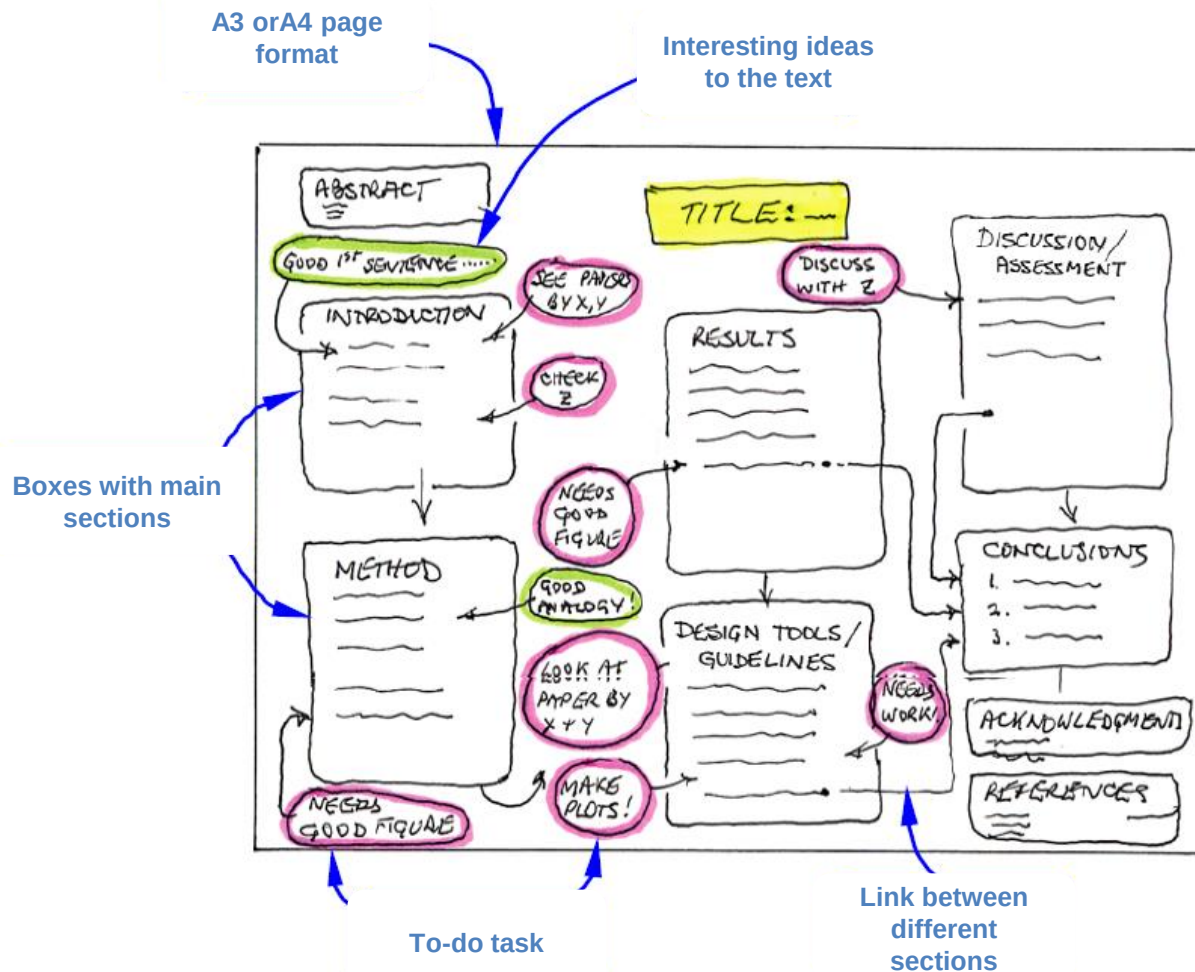
- ✓ Due to page limits, the extended abstract cannot show all results obtained in the dissertation
- ✓ The most relevant results must be carefully shown, together with the respective benchmarking (where possible, linked to the state-of-the-art presented in the introduction)

Conclusions

- ✓ Summary of main conclusions

References

- ✓ References in a language other than English must be translated into English. At the end, mention the original language - example: “(*in Portuguese*)”



And what follows?

